

Platform Architecture

Layer 1: Geospatial Intelligence Engine

The core differentiator: own this entirely

Component	Recommended Stack	Cost Model
GIS Backend	PostGIS (PostgreSQL extension)	Free / Open-source
Map Rendering	MapLibre GL JS (open-source Mapbox fork)	Free
Tile Server	Martin (Rust-based, PostGIS-native)	Free / Self-hosted
Basemap Tiles	OpenStreetMap + CARTO free tier	Free-\$50/mo
Isochrone Engine	OpenRouteService (self-hosted) or Valhalla	Free / Open-source
Satellite Imagery	ISRO Bhuvan API + Sentinel-2 (ESA, free)	Free
Spatial Analytics	GeoPandas + Shapely (Python)	Free
Flood / Risk Layers	NRSC Open Data + GHMC GIS portal	Free
Infrastructure Overlays	HMDA GIS Open Data + custom digitisation	Free data

 **Cost Insight:** MapLibre + PostGIS + self-hosted tiles replaces a ₹8–12L/year Mapbox enterprise licence. The savings alone justify the stack choice.

Layer 2: Digital Twin & XR Property Experiences

Component	Recommended Stack	Cost Model
3D Viewer (Web)	Three.js + React Three Fiber	Free
BIM Processing	IFC.js (open-source BIM viewer for web)	Free
360° Virtual Tours	Pannellum.js (open-source) or Marzipano	Free

Point Cloud Processing	Potree (open-source LiDAR renderer)	Free
AR (Mobile)	AR.js (web-based AR, no app download needed)	Free
3D Model Hosting	AWS S3 + CloudFront (CDN delivery)	~₹2–4/GB
Video Tour Streaming	Cloudflare Stream	\$5/1000 min
Property Photography	Matterport API (selective use for premium listings)	Pay-per-scan
Real-time Collaboration	Liveblocks (for multi-user XR viewing)	Free tier available

💡 **Cost Insight:** IFC.js + Three.js eliminates the need for Autodesk Forge (\$400+/month). Use Matterport only for ₹1Cr+ listings where scan cost (₹5–8K/property) is justified.

Layer 3: Legal Transparency Engine

Component	Recommended Stack	Cost Model
TGRERA Integration	REST scraping + official API (when available)	Free
Document OCR	Tesseract.js (open-source) + Google Document AI	Free + Pay-per-page
Document Storage	AWS S3 with server-side encryption	~₹2/GB/month
EC / Registration Lookup	Telangana Registration Dept API integration	Free (gov API)
Title Risk Scoring	Custom ML model (scikit-learn / XGBoost)	Free
Legal NLP	spaCy + custom fine-tuned model on Indian property law docs	Free
Blockchain Audit Trail	Polygon (low-cost EVM chain) for document hash anchoring	~\$0.001/tx

Compliance Monitoring	Apache Airflow (workflow scheduler for periodic checks)	Free / Self-hosted
Alert Engine	n8n (open-source automation)	Free / Self-hosted

💡 **Cost Insight:** Polygon's near-zero transaction fees make document hash anchoring viable at scale, immutable proof of listing document state without the cost of Ethereum mainnet.

Layer 4: AI-Powered Buyer Concierge

Component	Recommended Stack	Cost Model
LLM Backend	Claude API (Anthropic), Haiku for speed, Sonnet for depth	Pay-per-token
RAG Knowledge Base	LlamaIndex + pgvector (PostgreSQL)	Free
Embeddings	text-embedding-3-small (OpenAI) or Cohere Embed	~\$0.02/1M tokens
Conversational UI	Botpress (open-source) or custom React chat	Free
EMI / TCO Calculator	Custom Node.js service with RBI rate feed	Free
Recommendation Engine	Collaborative filtering, Surprise library (Python)	Free
Search Intelligence	Typesense (open-source Algolia alternative)	Free / Self-hosted
Scheduling Engine	Cal.com (open-source Calendly) self-hosted	Free
Feedback & NPS	Formbricks (open-source)	Free / Self-hosted
Analytics	PostHog (open-source product analytics)	Free self-hosted

💡 **Cost Insight:** Typesense replaces Algolia (₹15,000+/month at scale). Cal.com self-hosted replaces Calendly Business. PostHog replaces Mixpanel/Amplitude.

Layer 5: Core Platform Infrastructure

Component	Recommended Stack	Cost Model
Frontend	Next.js 14 (App Router) + TypeScript	Free
UI Component Library	shadcn/ui + Tailwind CSS	Free
Mobile App	React Native (Expo), single codebase for iOS + Android	Free
Backend API	FastAPI (Python), ideal for GIS + ML workloads	Free
API Gateway	Kong (open-source)	Free / Self-hosted
Database	PostgreSQL + PostGIS (primary), Redis (cache)	Free
Search	Typesense (self-hosted)	Free
Auth	Keycloak (open-source) or Supabase Auth	Free
File Storage	MinIO (self-hosted S3-compatible) or AWS S3	~₹2/GB
CDN	Cloudflare (generous free tier)	Free-\$20/mo
Containerisation	Docker + Kubernetes (K3s for cost-effective K8s)	Free
CI/CD	GitHub Actions	Free (2,000 min/mo)
Monitoring	Grafana + Prometheus + Loki (full stack)	Free / Self-hosted
Error Tracking	Sentry (open-source self-hosted)	Free

Cloud Provider	AWS India (ap-south-1, Mumbai region)	Pay-as-you-go
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Estimated Monthly Infrastructure Cost

Phase	Stage	Est. Monthly Cost
MVP	0–500 listings, <1,000 users	₹15,000 – ₹25,000
Growth	5,000 listings, 10,000 MAU	₹60,000 – ₹90,000
Scale	50,000+ listings, 100,000 MAU	₹2,00,000 – ₹3,50,000

These estimates cover AWS compute, storage, CDN, and third-party API costs. Geospatial and AI workloads are the primary cost drivers at scale, both are optimised by self-hosting open-source engines.

Recommended Development Team (Lean Build)

Role	Stack Ownership
Full-Stack Lead	Next.js, FastAPI, PostgreSQL
GIS Engineer	PostGIS, MapLibre, Python spatial stack
3D / XR Engineer	Three.js, IFC.js, React Three Fiber
ML / AI Engineer	LlamaIndex, RAG pipeline, recommendation engine
DevOps Engineer	K3s, GitHub Actions, AWS, Grafana

Stack Decision Summary

Principle	How It's Applied
Own the differentiators	GIS engine, Digital Twin viewer, Legal AI, all built in-house on open-source
Buy the commodities	Auth, email, payments, CDN, managed services only
Open-source first	Every proprietary SaaS has a free alternative chosen here

India-first hosting	AWS Mumbai region ensures data residency + low latency
Scale without re-architecture	PostGIS + pgvector + Redis is a stack that scales from MVP to 1M users without swapping engines